Cross-Lingual Representation Alignment by Token-Level Optimal Transport in a Language-Agnostic Space

Taisei Yamamoto[1,2], **Ryoma Kumon**[1,2], **Danushka Bollegala**[3], **Hitomi Yanaka**[1,2,4]
[1]The University of Tokyo [2]Riken [3]University of Liverpool [4]Tohoku University
{yamamo96, kumoryo9, hyanaka}@is.s.u-tokyo.ac.jp
danushka@liverpool.ac.uk

Abstract

Cross-lingual alignment (CLA) aims to align the representations of large language models (LLMs) across languages, enabling cross-lingual transfer to improve multilingual capabilities. Previous CLA methods often ignore language-specific information encoded in representations and only consider sentence-level alignment, which may lead to suboptimal performance and input-output language mismatch. We propose **CAROT** (**C**ross-Lingual **A**lignment of **R**epresentations in a Language-Agnostic Space via **O**ptimal **T**ransport), which consists of two steps: identifying language-specific representations in LLMs' internal states and aligning language-agnostic representations across languages at the token level by optimal transport, while explicitly preserving language-specific representations. Inference-time steering experiments show that the representations computed by CAROT are effective alignment targets, improving multilingual performance by up to 11.2 points in accuracy while maintaining input-output language consistency. We further use the representations obtained by CAROT as training targets, internalizing the aligned representations. The trained models outperform existing CLA methods in 11 of 18 evaluation settings (3 models $\times$ 3 tasks $\times$ ID/OOD languages). Our work provides insights into what constitutes effective alignment targets for CLA in LLMs. Code is available at https://github.com/ynklab/CAROT.

1 Introduction

As large language models (LLMs) are being used around the world, it is crucial that LLMs possess multilingual capabilities. However, previous studies have shown that LLMs have inconsistent abilities across languages, especially showing limited understanding of low-resource languages (e.g., Ahuja et al., 2023; Singh et al., 2025). As low-resource languages lack sufficient training data, it is important to achieve cross-lingual transfer, the ability of LLMs to generalize the knowledge and skills learned in one language to others. One key factor enabling cross-lingual transfer is cross-lingual alignment (CLA). CLA aligns the representations of LLMs across languages, allowing the model to map semantically similar inputs from different languages onto similar representations.

Previous CLA methods are mainly based on insights into the multilingual mechanisms of LLMs. Several studies have suggested that LLMs follow a three-stage pipeline when processing multilingual inputs: (1) mapping inputs into a semantic space, (2) performing inference in that semantic space, and (3) translating back into the output language for generation (Wu et al., 2025; Zhao et al., 2024; Tang et al., 2024). Multiple studies have reported a correlation between task performance in a given language and the similarity of its internal representations to those of the dominant language (Kargaran et al., 2025; Li et al., 2025). These findings have motivated the development of CLA methods aimed at increasing the cross-lingual similarity of the middle-layer representations of LLMs through inference-time steering (Zhao et al., 2025a; Li et al., 2026) and training (Liu and Niehues, 2025; Li et al., 2024; Zhao et al., 2025b).

To promote CLA by leveraging representations in high-resource languages and identifying ideal representations for multilingual inputs, we must address two problems: *what* to align and *where* to align it. Most training methods pull the hidden states of parallel sentences across languages closer together, but they do not necessarily consider the language-specific information encoded in the representations (Liu and Niehues, 2025; Li et al., 2024). Indiscriminately minimizing cross-lingual distance risks destroying the language-specific features models rely on, so language-specific components should be protected while aligning the language-agnostic components. Fur-

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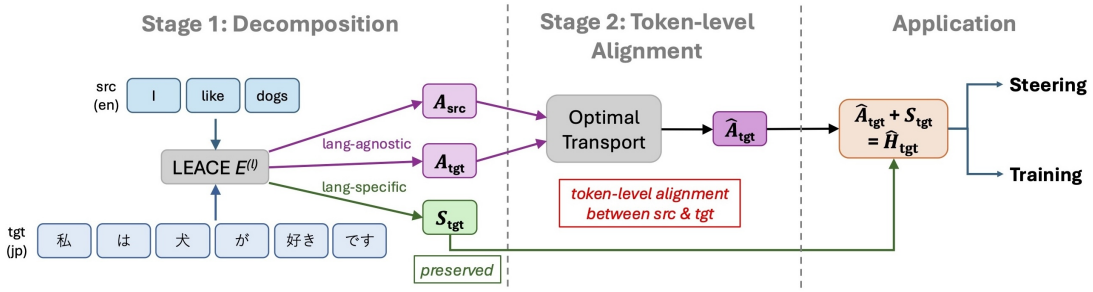


Figure 1: Overview of CAROT. **Stage 1 (Decomposition):** LEACE decomposes the hidden states of each token into language-agnostic components ($\mathbf{A}_{\text{src}}$ and $\mathbf{A}_{\text{tgt}}$) and a language-specific component ($\mathbf{S}_{\text{tgt}}$). **Stage 2 (Alignment):** Unbalanced optimal transport aligns $\mathbf{A}_{\text{src}}$ with $\mathbf{A}_{\text{tgt}}$ at the token level, producing ideal language-agnostic representations $\hat{\mathbf{A}}_{\text{tgt}}$. **Application:** The aligned representations are recombined with the preserved language-specific component to form $\hat{\mathbf{H}}_{\text{tgt}}$, which is used for inference-time steering or as a training signal.

thermore, many steering methods bring the multilingual representations closer to the model's central language by applying a global vector shift, but they do not perform token-level alignment (Zhao et al., 2025a; Li et al., 2026). As languages differ in morphological structure and grammatical constructions, a finer-grained token-level investigation of multilingual representations is necessary to develop more effective CLA methods. For example, *"I like dogs"* in English and *"Watashi wa inu ga suki desu"* in Japanese (Figure 1) have different word orders, and the Japanese sentence contains a topic marker "wa" and a subject marker "ga" that are not present in the English sentence.

To overcome these challenges, we propose **CAROT** (**C**ross-Lingual **A**lignment of **R**epresentations in a Language-Agnostic Space via **O**ptimal **T**ransport), a method to compute hidden representations that enhance the multilingual capabilities of LLMs by aligning token-level language-agnostic representations while preserving language-specific representations (Figure 1). We utilize LEACE (Belrose et al., 2023) to find linearly encoded language-specific information (§3.1) and isolate the language-agnostic representations. We then align the language-agnostic representations between the source and target languages at the token level via optimal transport (Chizat et al., 2018; Arase et al., 2023; Dou and Neubig, 2021), with dedicated adjustment for applying to LLMs' internal representations (§3.2). The source language is the dominant language in which the model performs best (e.g., English), and the target language is the language we want to improve. We confirm that the components removed by LEACE capture language-specific information and the remaining components correspond to language-agnostic semantic representations (§5). Our steering experiments, covering a broad range of tasks, demonstrate that the representations obtained by CAROT are effective alignment targets, significantly improving multilingual performance without inducing input-output language mismatch (§6).

Inference-time steering with CAROT, however, requires a source-language translation of the input, which makes it impractical to apply at scale. We therefore propose a training method that encourages the model's internal representations to match those obtained by CAROT (§7), so that translation is required only during training. Models trained with CAROT outperform existing CLA methods in 11 of 18 model-task-language settings (3 models $\times$ 3 tasks $\times$ ID/OOD languages), although the improvements do not yet match those observed with steering. Our work provides insights into the ideal multilingual representations and an interpretable framework for achieving CLA in LLMs, contributing to more effective cross-lingual transfer.

2 Related Work

Multilingual Mechanisms of LLMs. A number of studies have investigated the multilingual mechanisms of LLMs, revealing that LLMs process multilingual inputs through a shared semantic space in middle layers. Wendler et al. (2024) and Wu et al. (2025) applied logit lens to the middle layers of LLMs and found that the top-ranked language is often the dominant language of the model. Chang et al. (2022) demonstrated that languages share common linear subspaces in the middle layers of multilingual LLMs. These findings support the hypothesis that LLMs first map multilingual inputs into a shared semantic space and then perform inference in that space.

Cross-Lingual Alignment (CLA). In addition to the multilingual mechanisms of LLMs described above, Kargaran et al. (2025) and Li et al. (2025) demonstrated that LLMs perform better in a language for which the hidden representations are closer to those of the dominant language. Based on these findings, several studies applied inference-time steering to bring the multilingual representations closer to those of the dominant language (Zhao et al., 2025a; Lim et al., 2025). Furthermore, Liu and Niehues (2025) and Li et al. (2024) proposed contrastive training methods to increase the similarity of the middle-layer representations of LLMs across languages. However, these methods only consider sentence-level alignment, and most training methods ignore language-specific information encoded in representations. Although Zhao et al. (2025b) proposed a cross-lingual training method that took into account language-specific representations, they only calculated and aligned sentence-level representations. Token-level cross-lingual alignment has also been studied for word alignment, with several methods using optimal transport (Dou and Neubig, 2021; Arase et al., 2023), but not for CLA in LLMs. Our approach addresses both issues by aligning token-level language-agnostic representations.

Concept Erasure. Concept erasure identifies and removes specific information from representations without compromising other features (Ravfogel et al., 2020, 2022). Belrose et al. (2023) developed LEACE, a concept erasure method that computes an affine transformation that removes the linear features representing the target concept (Appendix B). They provided theoretical guarantees that after applying LEACE, the target labels cannot be classified linearly while minimizing the distance between the original and transformed representations. In the multilingual setting, earlier work removed language-specific components from multilingual encoders in a predefined form, by subtracting per-language centroids (Libovický et al., 2020), normalizing language-specific means and variances (Zhao et al., 2021), or projecting out low-rank language subspaces (Xie et al., 2022; Zhao et al., 2025a). We utilize LEACE to isolate language-specific features from the internal representations of LLMs.

3 CAROT

We propose CAROT, a method for computing multilingual representations of LLMs to enhance CLA. CAROT consists of two stages: (1) decomposing hidden representations into language-specific and language-agnostic components using LEACE (§3.1), and (2) aligning the language-agnostic representations of the source and target languages at the token level via optimal transport (§3.2).

3.1 Language-Specific and Language-Agnostic Decomposition

Sentence Representations. To identify features encoding language information in LLMs' internal representations, we calculate sentence-level embeddings and train LEACE on these representations. Specifically, we calculate the sentence representation $\mathbf{x}^{(l)}$ (at layer l) by aggregating the hidden states $\mathbf{h}_t^{(l)}$ using a weighted average across token positions t, following Muennighoff (2022):

$$\mathbf{x}^{(l)} = \sum_{t=1}^{n} w_t \mathbf{h}_t^{(l)}, \quad \text{where} \quad w_t = \frac{t}{\sum_{i=1}^{n} i} \quad (1)$$

and n is the number of tokens in the sentence. This aggregation averages out token-level variation to expose the dominant semantic content across parallel sentence pairs, leaving the language direction more salient for LEACE to identify.

LEACE-Based Erasure. As described in §2, LEACE calculates an affine transformation after which the target labels cannot be linearly separated. We train LEACE in a multiclass setting, where the target labels are the languages of the input sentences. LEACE is trained at each layer l, denoted as a function $E^{(l)}$. LEACE is subsequently applied to the hidden vectors of each token.

3.2 Token-Level Alignment with Optimal Transport in a Language-Agnostic Space

CAROT aligns the token-level language-agnostic representations of the target language with those of the source language, while preserving the language-specific representations of the target language. We select English as the source language as it is the dominant language for all models in our experiments and thus serves as a good anchor for alignment.

LEACE-Based Decomposition. Let $\mathbf{H}_{\text{src}} \in \mathbb{R}^{n_{\text{src}} \times d}$ and $\mathbf{H}_{\text{tgt}} \in \mathbb{R}^{n_{\text{tgt}} \times d}$ be the hidden states of the parallel sentences at layer l, where n_{src} and n_{tgt} are the number of tokens in the source and target language sentences, respectively, and d is the hidden dimension. We first apply the LEACE eraser

$E^{(l)}$ to separate language-specific and language-agnostic representations:

$$\mathbf{A}_{\text{src}} = E^{(l)}(\mathbf{H}_{\text{src}}),\ \mathbf{A}_{\text{tgt}} = E^{(l)}(\mathbf{H}_{\text{tgt}}), \quad (2)$$

$$\mathbf{S}_{\text{tgt}} = \mathbf{H}_{\text{tgt}} - \mathbf{A}_{\text{tgt}}, \quad (3)$$

where $E^{(l)}$ is applied to each row of $\mathbf{H}$, i.e., each token's hidden vector. $\mathbf{A}$ denotes the language-agnostic representation, and $\mathbf{S}_{\text{tgt}}$ is the language-specific component of the target language.

Optimal Transport in Language-Agnostic Space. We align source and target language-agnostic representations via optimal transport (OT), using Mahalanobis distance as the transport cost. We precompute the covariance matrix $\mathbf{\Sigma}$ of sentence representations (Equation 1) on the same training data as LEACE. We define the cost matrix $\mathbf{C} \in \mathbb{R}^{n_{\text{tgt}} \times n_{\text{src}}}$ as the squared Mahalanobis distance:

$$C_{ij} = \left\| \mathbf{\Sigma}^{-1/2}(\mathbf{a}_{\text{tgt},i} - \mathbf{a}_{\text{src},j}) \right\|_2^2, \quad (4)$$

where $\mathbf{a}_{\text{tgt},i}^\top$ and $\mathbf{a}_{\text{src},j}^\top$ are the i-th and j-th row of $\mathbf{A}_{\text{tgt}}$ and $\mathbf{A}_{\text{src}}$, which correspond to the language-agnostic representations of the i-th target and j-th source token, respectively. We use Mahalanobis distance considering that the hidden vectors of LLMs tend to be anisotropic[1] (Machina and Mercer, 2024).

Since the source and target sequences generally differ in length ($n_{\text{src}} \neq n_{\text{tgt}}$) and not every token has a clear cross-lingual counterpart, we solve the *unbalanced* OT problem with the Sinkhorn algorithm[2] (Chizat et al., 2018; Flamary et al., 2021) under uniform marginal distributions $\mathbf{p} = \frac{1}{n_{\text{tgt}}}\mathbf{1}$ and $\mathbf{q} = \frac{1}{n_{\text{src}}}\mathbf{1}$:

$$\mathbf{T} = \text{Sinkhorn-UB}(\mathbf{p}, \mathbf{q}, \mathbf{C};\ \epsilon_{\text{OT}},\ \rho), \quad (5)$$

where ϵ_{OT} controls entropic regularization and ρ relaxes the marginal constraints.

Perplexity-Based Filtering. The transport plan $\mathbf{T}$ may assign diffuse mass to tokens that are difficult to align. To suppress such noise, we compute the perplexity of the row-normalized transport distribution for each target token i:

$$\text{ppl}_i^{\text{row}} = \exp\Big(-\sum_j \hat{T}_{ij} \log \hat{T}_{ij}\Big), \quad (6)$$

$$\text{where} \quad \hat{T}_{ij} = \frac{T_{ij}}{\sum_j T_{ij} + \epsilon_{\text{num}}},$$

and analogously for columns (source tokens). ϵ_{num} is a small constant for numerical stability. Rows with $\text{ppl}_i^{\text{row}} > \theta \cdot n_{\text{src}}$ and columns with $\text{ppl}_j^{\text{col}} > \theta \cdot n_{\text{tgt}}$ are zeroed out, where θ is a hyperparameter. This filtering retains only those target tokens that are confidently matched to a focused subset of source tokens, and vice versa.

Transported Representation. For each target token i, the received transport mass $m_i = \sum_j T_{ij}$ quantifies alignment confidence. We compute the new language-agnostic representation as a soft interpolation between the transported source and the original target:

$$\hat{\mathbf{A}}_{\text{tgt}} = \boldsymbol{\alpha} \odot \hat{\mathbf{T}}\mathbf{A}_{\text{src}} + (\mathbf{1} - \boldsymbol{\alpha}) \odot \mathbf{A}_{\text{tgt}}, \quad (7)$$

where $\hat{\mathbf{T}}$ is the row-normalized filtered transport plan and $\alpha_i = \min(1, m_i/p_i)$ scales each target token's update proportionally to its received mass. Tokens with little or no received mass (e.g., by being filtered out) thus retain their original language-agnostic representation.

Language-Specific Component Restoration. Finally, we add back the preserved target language-specific component:

$$\hat{\mathbf{H}}_{\text{tgt}} = \hat{\mathbf{A}}_{\text{tgt}} + \mathbf{S}_{\text{tgt}}. \quad (8)$$

The resulting $\hat{\mathbf{H}}_{\text{tgt}}$ carries language-agnostic representations aligned with the source language while retaining the target language identity.

3.3 Using CAROT Representations

We use $\hat{\mathbf{A}}_{\text{tgt}}$ and $\hat{\mathbf{H}}_{\text{tgt}}$ in two ways: as (1) an inference-time steering method and (2) a training target to encourage the model to produce better-aligned representations.

Inference-Time Application. When used for inference-time steering, CAROT is applied to the hidden states of the tokens corresponding to a question (and answer options for multiple-choice tasks) at a single layer, swapping in the new representations $\hat{\mathbf{H}}_{\text{tgt}}$ before the model continues forward. BOS, EOS, and padding tokens are excluded from steering to preserve the model's generation behavior. Note that steering requires a source-language translation of the input. Our steering evaluation (§6) therefore serves to demonstrate the effectiveness of the representations computed by CAROT as alignment targets, while the approach itself remains applicable to offline settings where parallel inputs

[1] We confirm that especially gemma3-4b has highly anisotropic distributions of hidden representations, as discussed in Appendix C.

[2] https://github.com/PythonOT/POT

are naturally available, such as generating higher-quality reasoning traces in low-resource languages as distillation data.

Training Objective. When used for training, we fine-tune the model with two loss components: language-agnostic alignment loss $\mathcal{L}_{\text{align}}$ and language modeling loss $\mathcal{L}_{\text{LM}}$. *Language-agnostic alignment loss* is calculated on parallel data. We penalize the deviation of the model's current language-agnostic component $\mathbf{A}_{\text{tgt}}$ from the CAROT-derived $\hat{\mathbf{A}}_{\text{tgt}}$ using the same Mahalanobis distance as OT cost:

$$\mathcal{L}_{\text{align}} = \frac{1}{n_{\text{tgt}}} \sum_{i=1}^{n_{\text{tgt}}} \frac{\left\| \boldsymbol{\Sigma}^{-1/2} \left(\mathbf{a}_{\text{tgt},i} - \hat{\mathbf{a}}_{\text{tgt},i} \right) \right\|_2^2}{\left\| \boldsymbol{\Sigma}^{-1/2} \hat{\mathbf{a}}_{\text{tgt},i} \right\|_2^2 + \epsilon_{\text{num}}}. \quad (9)$$

Normalizing the loss by the target norm stabilizes the magnitude of the loss. *Language modeling loss* is calculated on the instruction-tuning data. We minimize the supervised fine-tuning (SFT) loss $\mathcal{L}_{\text{LM}}$ on the answer tokens of the target batch. In the first half of training, we optimize the two losses alternately at each training step rather than combining them into a single loss to avoid interference between the two objectives. In the second half, we only optimize $\mathcal{L}_{\text{LM}}$ to allow the model to leverage its aligned representations for generation.

Unlike steering, this application requires parallel data only during training. As parallel corpora are available for many languages and our training method requires only a modest amount of parallel data, training with CAROT is a practical route to improving the multilingual ability of LLMs.

4 Experimental Setup

Models and Datasets. In our experiments, we use three multilingual LLMs: *Llama3.1-8B* (Grattafiori et al., 2024), *gemma3-4b* (Gemma Team et al., 2025), and *Qwen3-4B* (Qwen Team, 2025). For training LEACE, we use parallel sentences from two multilingual datasets: WMT24++ (Deutsch et al., 2025) and NTREX (Federmann et al., 2022). We include 12 in-distribution (ID) languages for LEACE, which vary in terms of their linguistic characteristics and resource levels (see Table 4). In total, we use 2,560 sentences per language for training LEACE.

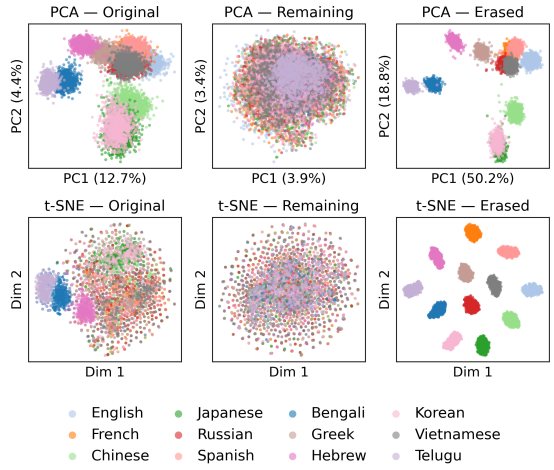
For evaluation, we use KLAR Continent category (Wang et al., 2025) as the knowledge retrieval task, GMMLU (Singh et al., 2025) *test* set as the task that requires both knowledge and reasoning, BELEBELE (Bandarkar et al., 2024) for reading comprehension, and XQuAD (Artetxe et al., 2020) for open-ended question answering. We also use FLORES+ (NLLB Team et al., 2024) for analyzing the internal representations of the models in §5. The source language is English, and the target languages are seven languages for KLAR, nine for GMMLU and BELEBELE, and five for XQuAD, selected based on the availability of the datasets and the variety of characteristics (Table 4).

Hyperparameter Search. In the steering and training experiments in §6 and §7, we perform a grid search to find the optimal hyperparameters for each method and model. We use the *dev* set of GMMLU and adopt the parameters achieving the best accuracy. For steering with CAROT, we search for the best layer from every three layers, and the filtering threshold θ from $\{0.25, 0.5, 0.75, 1.0\}$ (note that θ=1 means that no filter is applied). For training with CAROT, we adopt the best layer and filtering threshold from the steering experiments for each model, and search for the best learning rate from {5e-4, 1e-4, 5e-5, 1e-5}.

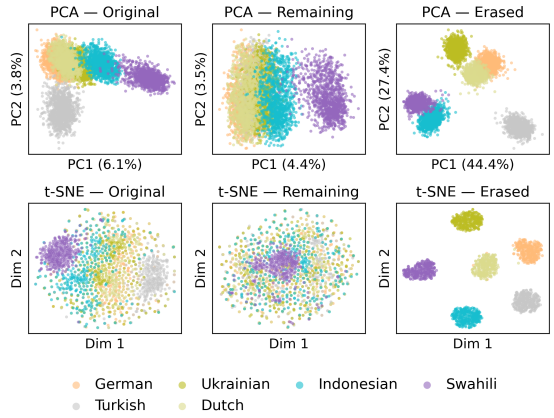
Metrics. We use accuracy, (language) fidelity, and cross-lingual agreement ratio as evaluation metrics. Fidelity is the percentage of instances where the model's answer is in the same language as the input. We use GlotLID (Kargaran et al., 2023) to identify the language of the model's output. Cross-lingual agreement ratio is the percentage of instances for which the model's answer in the target language matches that in the source language. The details of the evaluation protocols are described in Appendix E.

5 Analysis of Language-Specific and Language-Agnostic Representations

First, we analyze the representations obtained by applying LEACE on sentence-level embeddings (§3.1). Figure 2 shows PCA-based and t-SNE-based (van der Maaten and Hinton, 2008) 2D scatter plots of the original, remaining, and LEACE-erased sentence representations of Llama3.1-8B at layer 12, calculated on the FLORES+ *devtest* set. We show the results for 12 ID languages and 6 out-of-distribution (OOD) languages, which are not used for training LEACE. For ID languages, sentences in the remaining representations are mixed across languages, while the LEACE-erased representations are clearly separated by language. These



(a) In-distribution languages



(b) Out-of-distribution languages

Figure 2: Scatter plot of original (left), remaining (center), and LEACE-erased (right) representations of Llama3.1-8B at layer 12. Each point represents the sentence embedding, colored by language.

observations confirm that LEACE successfully identifies language-specific representations. For OOD languages, the remaining representations are somewhat clustered by language, indicating residual language information not captured by LEACE. Meanwhile, the erased representations still show clear separation. This suggests that LEACE captures language-specific representations that generalize to OOD languages to some extent. The same trend holds across other layers and models (Appendix C).

Figure 3 shows the average cosine similarity of the sentence representations between the intra-language pairs (i.e., sentences in the same language) and inter-language pairs (i.e., parallel sentence pairs between the source and target languages) for Llama3.1-8B, calculated on the FLORES+ *devtest* set. In the remaining representations, inter-language similarity is higher than intra-language similarity, showing that content rather than language determines the representation (i.e., the representations are language-agnostic). In contrast, in the LEACE-erased representations, intra-language similarity is consistently close to 1, indicating that erased representations follow a common direction within the same language regardless of content (i.e., the representations are language-specific). These results confirm that LEACE successfully disentangles language-specific and language-agnostic components.

6 Steering Evaluation

6.1 Baselines

We implement two steering methods for comparison with CAROT. One is a vector offset based on diff-in-means (Lim et al., 2025), which we refer to as *Lang Vec*. The other is *MLRS* (Zhao et al., 2025a), which first identifies language-specific components and then suppresses them in the middle layers and adds them back in the later layers. Note that these methods perform sentence-level alignment, while CAROT is a token-level alignment method. Also note that although MLRS considers language-specific representations, it does not give theoretical grounds for the decomposition, while CAROT does by using LEACE. We describe the details of these baselines in Appendix F.

6.2 Results and Analysis

Table 1 shows the task accuracy and language fidelity for each method, averaged across target languages. CAROT outperforms the baselines in most cases, achieving a significant improvement of up to 11.2 points in accuracy over the original model (Unsteered). Meanwhile, fidelity does not degrade substantially, indicating that CAROT does not cause input-output language mismatch. For gemma3-4b on GMMLU, fidelity is lower than in the other settings for all methods, which may be due to the model's tendency to generate reasoning steps in English even when the input is in another language. Additionally, Table 7 in Appendix H reports the agreement ratio, showing that CAROT achieves the highest agreement in most cases. These results indicate that CAROT improves CLA.

The best steering layers of CAROT are all in the early-to-middle layers (layer 12, 6, 9 for Llama3.1-8B, gemma3-4b, and Qwen3-4B, respectively, as shown in Appendix G). As the intervention targets

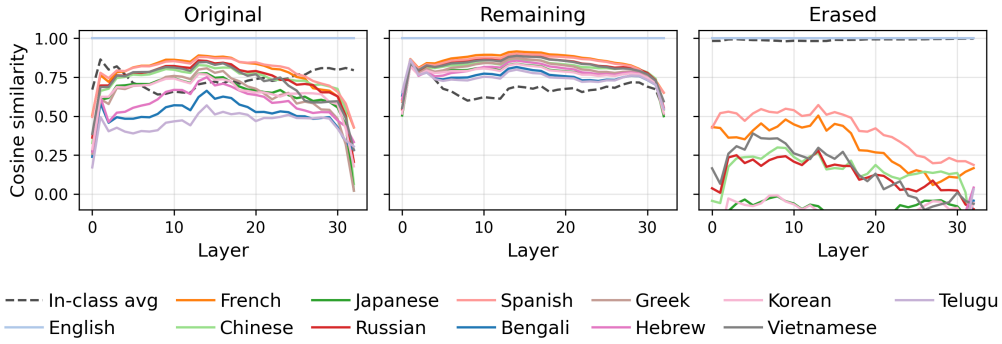


Figure 3: Cosine similarity between the intra-language (dashed, averaged across languages) and inter-language (solid) sentence pairs in the original, remaining, and LEACE-erased representations of Llama3.1-8B across layers.

		GMMLU (9)		KLAR (7)		BELEBELE (9)		XQuAD (5)	
Model	Method	Acc	Fid	Acc	Fid	Acc	Fid	Acc	Fid
Llama3.1-8B	Unsteered	48.4	93.5	68.4	94.7	74.7	87.7	35.0	77.9
	Lang Vec	43.8	93.0	69.6†	94.5	74.5	80.7	37.2†	76.8
	MLRS	43.7	93.8	69.5†	94.6	77.5†	88.8	35.6†	78.0
	CAROT (Ours)	**49.8**†	93.1	**73.3**†	93.8	**81.9**†	85.3	**41.2**†	72.7
gemma3-4b	Unsteered	49.4	64.7	67.5	92.7	64.2	94.4	21.9	85.9
	Lang Vec	49.0	63.9	67.9	92.8	63.8	94.1	21.8	86.3
	MLRS	48.2	66.8	67.2	92.0	64.8	95.1	23.5†	85.1
	CAROT (Ours)	**51.8**†	62.6	**75.8**†	96.4	**66.0**†	92.9	**26.0**†	82.5
Qwen3-4B	Unsteered	56.8	98.5	56.8	93.7	62.3	90.1	45.5	72.0
	Lang Vec	54.9	97.0	59.1†	93.9	63.3†	75.8	**45.8**	68.9
	MLRS	55.9	98.5	59.2†	93.9	**64.9**†	89.2	45.6	71.6
	CAROT (Ours)	**59.6**†	98.1	**68.0**†	90.4	62.5	89.5	45.6	69.9

Table 1: Steering results averaged across target languages. Each row reports task accuracy and fidelity (%) of four methods: **Unsteered**, **Lang Vec**, **MLRS**, and **CAROT** (Ours). The source language is English for all cases, and the number of target languages is shown in parentheses for each dataset. Bold marks the highest value per metric group. † indicates statistically significant accuracy improvement over Unsteered ($p < 0.05$, one-sided McNemar test across all languages).

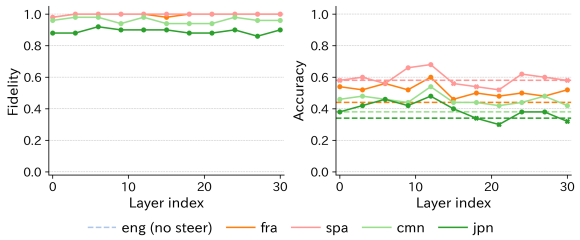


Figure 4: Fidelity (left) and accuracy (right) of Llama3.1-8B on GMMLU dev calculated when conducting hyperparameter search for the best steering layer with CAROT. For other parameters (e.g., θ), we use the best values at each layer. Dashed lines show the original model (Unsteered) performance.

only the question tokens, steering applied at later layers may have a weaker effect once the relevant information has already been propagated to the answer-generating position.

Figure 4 shows layer-wise per-language fidelity and accuracy for Llama3.1-8B on the GMMLU *dev* set when steering with CAROT, calculated when conducting hyperparameter search. CAROT achieves high fidelity across all languages and layers, and the accuracy improves most in the early layers. In contrast, the results for Lang Vec show degraded fidelity in many cases (Figure 11). Lang Vec brings the hidden states of the target language closer to those of the source language, which can cause a loss of language-specific information and thus low fidelity. Additional results are shown in Appendix H.

Figure 5 shows the cosine similarity of language-agnostic sentence representations between the source and target languages when steering with CAROT at layer 12 of Llama3.1-8B, calculated on the FLORES+ *devtest* set. Similarity increases at the steering layer, and higher similarity lasts in later

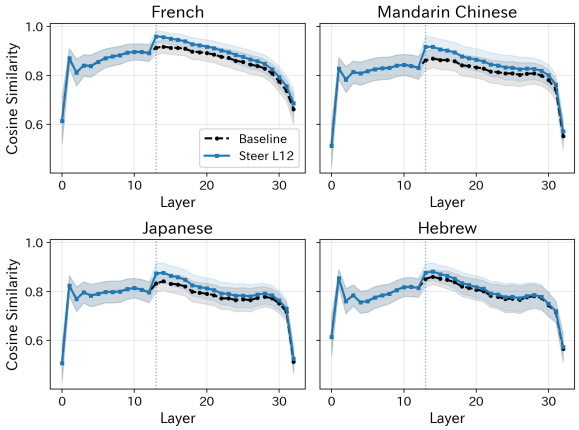


Figure 5: Cosine similarity of the language-agnostic components of sentence representations between the source and target languages when applying CAROT steering at layer 12 of Llama3.1-8B.

layers. This confirms that CAROT successfully aligns the language-agnostic representations between the source and target languages. As a sanity check, we inspect several samples and confirm that transport plans are sparse and exhibit little noise, and that semantically corresponding tokens are correctly aligned across languages (Appendix D).

An ablation study (Table 8, Appendix H) shows that LEACE-based decomposition and filtering of the OT plan contribute to the robustness of CAROT. Removing LEACE-based decomposition degrades performance in most cases. Removing the filtering yields comparable accuracy but is less stable (e.g., a drop of 14.0 points on Qwen3-4B/KLAR) and lowers fidelity in many settings. In addition, we compare CAROT with a variant that swaps the language-agnostic representations of only the last token between the source and target languages to investigate the effect of token-level alignment. The results show that CAROT outperforms this variant in most cases, indicating that token-level alignment effectively improves accuracy.

The computational complexity of CAROT is $O(n_{\text{src}} n_{\text{tgt}}(d + T))$, where T is the number of iterations for Sinkhorn's algorithm (we set $T = 1000$). Table 2 shows the average per-forward-pass latency for Llama3.1-8B on the GMMLU *test* set (30 samples per language). CAROT adds a second, source-language forward pass and the OT computation, roughly doubling the prefill cost (e.g., $24.3 + 29.6 = 53.9$ ms vs. 28.1 ms for Spanish). However, since CAROT is applied only once at prefill, this cost is amortized over the entire generation process, which involves as many forward passes as the number of tokens generated. The overhead of CAROT is therefore modest in practice, especially for long generations.

Lang	orig fwd	src fwd	CAROT fwd
Spanish	28.1 ± 2.6	24.3 ± 0.8	29.6 ± 4.1
Japanese	28.7 ± 2.7	28.2 ± 2.9	30.8 ± 6.3
Telugu	36.8 ± 15.8	24.4 ± 1.8	49.6 ± 30.0

Table 2: Computational overhead of CAROT measured on Llama3.1-8B with 30 samples from the GMMLU *test* set. We show the average time (in milliseconds) for one forward pass of the original model in the target language (*orig fwd*), the original model in the source language (*src fwd*), and the model steered with CAROT (*CAROT fwd*). The average number of tokens in the source language (English) is 83.1, and in the target languages, 122.2 (Spanish), 122.1 (Japanese), and 601.4 (Telugu).

7 Training Experiments

Finally, we evaluate the training application of CAROT described in §3.3, where the model is trained to produce language-agnostic representations that match those calculated by CAROT.

7.1 Training Settings

To assess the effectiveness of CAROT in post-training, we apply instruction-tuning-style training to the base (i.e., non-instruction-tuned) models of Llama3.1-8B, gemma3-4b, and Qwen3-4B, incorporating cross-lingual alignment objectives. For the alignment data to calculate $\mathcal{L}_{\text{align}}$, we use WMT24++ and NTREX. For the instruction-tuning data to calculate $\mathcal{L}_{\text{LM}}$, we use Aya Dataset (Singh et al., 2024), a multilingual instruction-tuning dataset. We train the models in five high-resource languages and evaluate in both ID and OOD languages, including low-resource languages (Table 4), reflecting the practical scenario where massive training data is available in only high-resource languages. The evaluation datasets are the same as described in §4, except that we omit XQuAD. Unlike the other three datasets, XQuAD requires extractive generation, and the models instruction-tuned in our setting fail to consistently produce answers in the required format under all settings, making the resulting accuracy uninformative.

We compare CAROT with standard SFT (*LM*) and *MidAlign* (Liu and Niehues, 2025), which minimizes the contrastive loss to increase the similarity of the middle-layer representations on the parallel sentences across languages. We fine-tune the mod-

Model	Method	GMMLU ID	GMMLU OOD	KLAR ID	KLAR OOD	BELEBELE ID	BELEBELE OOD
Llama3.1-8B	LM	43.2 ± 6.3	31.4 ± 5.6	61.7 ± 11.8	54.3 ± 13.1	61.1 ± 15.4	46.3 ± 5.7
	MidAlign	45.3 ± 4.9	**34.5** ± 5.9	63.1 ± 13.8	47.0 ± 15.0	65.3 ± 13.2	42.3 ± 9.2
	CAROT (Ours)	**46.5** ± 4.8	31.8 ± 5.3	**66.9** ± 9.4	**56.1** ± 12.0	**66.5** ± 15.7	**51.5** ± 8.5
gemma3-4b	LM	33.5 ± 3.2	30.5 ± 2.3	54.3 ± 12.3	38.7 ± 4.2	50.3 ± 6.9	40.8 ± 4.2
	MidAlign	**34.8** ± 5.5	**32.0** ± 1.4	56.4 ± 15.0	41.2 ± 1.4	**60.3** ± 10.7	**50.5** ± 6.0
	CAROT (Ours)	34.7 ± 5.9	31.5 ± 2.7	**57.5** ± 14.1	**42.9** ± 3.8	57.8 ± 8.6	49.8 ± 4.6
Qwen3-4B	LM	64.6 ± 4.5	49.4 ± 5.6	71.2 ± 15.1	**51.6** ± 12.3	79.8 ± 4.6	**68.9** ± 6.5
	MidAlign	65.3 ± 4.4	48.7 ± 4.1	74.6 ± 13.7	40.9 ± 17.7	76.6 ± 10.2	65.7 ± 4.9
	CAROT (Ours)	**66.1** ± 4.0	**49.6** ± 5.8	**75.2** ± 10.5	45.0 ± 15.7	**81.9** ± 8.5	68.4 ± 6.0

Table 3: Accuracy (%) on GMMLU, KLAR, and BELEBELE, aggregating ID and OOD languages. Values are mean ± standard deviation across languages within each group. Here ID/OOD indicates whether the language is included in the fine-tuning data. Bold marks the highest mean within each ID/OOD group.

els with LoRA (Hu et al., 2022) in all settings, and further details are described in Appendix I.

7.2 Results and Analysis

Table 3 shows the evaluation results. CAROT achieves the best accuracy in 11 out of 18 model-task-language settings (3 models × 3 tasks × ID/OOD languages), indicating that it is effective in improving multilingual performance when incorporated into post-training. To analyze the effect of our training method on the internal representations, we calculate the cosine similarity between the source and target language-agnostic sentence representations before and after training on FLORES+ *devtest* set. The results for Llama3.1-8B are shown in Figure 12 (Appendix I). We confirm that language-agnostic representations are more aligned after training with CAROT, indicating the intended effect of our training objective.

However, the overall performance gains by training are smaller than those by steering (§6). One possible reason is that the LEACE erasers and covariance matrices are precomputed and fixed during training, but the model's representations change, which may reduce the effectiveness of the alignment loss. To quantify this effect, we refit LEACE and the covariance matrix on the post-training representations of Llama3.1-8B at layer 12, which is the layer where the alignment loss is applied, and measure principal angles between the subspaces spanned by the top 50 eigenvectors of the pre- and post-training matrices. The averaged principal angles are 40.3° (max. 75.1°) for LEACE and 32.8° (max. 87.1°) for the covariance matrix. These large angles indicate a representation drift, which may make the precomputed parameters increasingly inaccurate. One possible remedy is periodically updating the LEACE and covariance matrices during training, which we leave for future work.

Moreover, our training with CAROT yields smaller improvements in GMMLU, which includes reasoning-heavy problems. Although our experiments only consider SFT-based objectives, recent training approaches often adopt reinforcement learning (RL) to improve reasoning capabilities, such as GRPO (Shao et al., 2024). Incorporating CAROT into RL-based training is an interesting direction for future work.

8 Conclusion

We present CAROT, which disentangles language-specific and language-agnostic components in LLMs' hidden states and aligns language-agnostic representations across languages at the token level while preserving language-specific representations. Our steering experiments show that representations calculated by CAROT enhance the multilingual performance of LLMs without compromising input-output language consistency. As steering requires a source-language translation of the input and is thus impractical to apply at scale, we propose to use representations obtained by CAROT as training objectives, so that translation is needed only during training. We demonstrate that models trained with CAROT outperform previous CLA methods in 11 of 18 evaluation settings, although the gains do not fully match those observed with steering. Our work provides insights into what constitutes effective alignment targets for CLA in LLMs.

Limitations

Our experiments are conducted on 4–8B parameter models due to computational resource constraints.

As Lim et al. (2025) suggest that the multilingual mechanisms of LLMs may differ by model size, the effectiveness of our method on larger models remains to be investigated.

In CAROT, we use LEACE to identify language-specific and language-agnostic components in the model's representations. LEACE considers only linear features, while LLMs involve non-linear transformations. Thus, it is possible that CAROT may overlook non-linear language features that LLMs actually use, which may limit the effectiveness of our method.

As discussed in §7.2, our training relies on SFT-based objectives, while recent training approaches usually adopt RL to improve reasoning capabilities. We expect that CAROT can be incorporated into RL frameworks by alternately applying $\mathcal{L}_{\text{align}}$ and the RL loss at each training step, which is left for future work.

Additionally, as discussed in §7.2, we use the precomputed LEACE and covariance matrices obtained from the base model throughout training. However, the model's representations drift during training, as shown by the large principal angles between the pre- and post-training LEACE and covariance matrices. Although we confirm that language-agnostic representations are more aligned between the source and target languages after training (§7.2), periodically updating the LEACE and covariance matrices during training may provide a more accurate training signal, which is also left for future work.

Han et al. (2025) point out that cross-lingual alignment often sacrifices the model's ability to answer culturally specific questions, as representations are aligned to the dominant language. Meanwhile, CAROT preserves language-specific representations, which may help mitigate this issue. We do not evaluate the model's cultural understanding in this work, and leave it for future work.

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A Details on Models and Datasets

All the model weights used in our experiments are obtained via Hugging Face Hub, and the datasets are obtained from Hugging Face Hub or the official websites of the datasets (e.g., GitHub repositories). In the steering experiment (§6), we use three instruction-tuned multilingual LLMs: *Llama3.1-8B*[3], *gemma3-4b*[4], and *Qwen3-4B*[5]. In the training experiment (§7), we use the base (i.e., non-instruction-tuned) variants of the three models. In §4, we train LEACE using parallel sentences from two datasets: WMT24++[6] and NTREX[7]. We also use FLORES+[8] for analysis of the language-specific and language-agnostic representations (§5). In §6 and §7, we use KLAR Continent category[9], Global-MMLU (GMMLU) *test* set[10], BELEBELE[11], and XQuAD[12] for evaluation. As an instruction-tuning dataset, we use Aya Dataset[13]. We list the languages used in each experiment in Table 4.

B LEACE

LEACE (Belrose et al., 2023) is a method for concept erasure that calculates an affine transformation to erase a concept from vectors while preserving the remaining information. Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be a matrix of n vectors of dimension d and $\mathbf{Z} \in \mathbb{R}^{n \times k}$ be the one-hot encoded concept labels over k classes. They first calculate the covariance matrix $\mathbf{\Sigma}_{\mathrm{XX}}$ and the cross-covariance matrix $\mathbf{\Sigma}_{\mathrm{XZ}}$. Here, let $\mathbf{W} = (\mathbf{\Sigma}_{\mathrm{XX}}^{\frac{1}{2}})^{+}$ and $\mathbf{P}_{\mathbf{W}\mathbf{\Sigma}_{\mathrm{XZ}}} = (\mathbf{W}\mathbf{\Sigma}_{\mathrm{XZ}})(\mathbf{W}\mathbf{\Sigma}_{\mathrm{XZ}})^{+}$. Then, the LEACE transformation E is defined as:

$$E(\mathbf{x}) = \mathbf{P}\mathbf{x} + \mathbb{E}[\mathbf{x}] - \mathbf{P}\mathbb{E}[\mathbf{x}], \tag{10}$$

where $\mathbf{P} = \mathbf{I} - \mathbf{W}^{+}\mathbf{P}_{\mathbf{W}\mathbf{\Sigma}_{\mathrm{XZ}}}\mathbf{W}$ and $\mathbb{E}[\mathbf{x}]$ is the mean of $\mathbf{X}$. This transformation ensures that no linear classifier can predict $\mathbf{Z}$ from $E(\mathbf{X})$ better than a constant predictor, while minimizing the expected squared distance between the original and transformed vectors. We apply LEACE to hidden states $\mathbf{h}$ of LLMs with the language of the input sentence as the concept $\mathbf{Z}$, so that $E(\mathbf{h})$ gives the language-agnostic component and $\mathbf{h} - E(\mathbf{h})$ the language-specific one.

C Additional Results for Language-Specific and Language-Agnostic Representations

In §5, we show the 2D scatter plots of the original, remaining, and LEACE-erased representations of Llama3.1-8B at layer 12. Here, we show the scatter plots of Llama3.1-8B at layers 6 and 25 in Figure 6. We can observe that the same trend as layer 12 holds at both layers, with the LEACE-erased representations showing clearer separation by language, while the remaining representations are mixed together regardless of language.

We also show the scatter plots of gemma3-4b and Qwen3-4B at layers 6 and 7 in Figure 7. For gemma3-4b, the original representations are more mixed across languages in the PCA-based visualization compared to other models. This can be due to anisotropic distribution of the hidden states of gemma3-4b, with a large contribution ratio of the top principal component (PC) (we observe 89.6% contribution ratio of the top PC at layer 13). Nonetheless, LEACE successfully identifies language representations, as shown by the similar trend in the scatter plots.

Finally, we train a linear classifier to identify the language of the input sentence on the representations on the FLORES+ *dev* set and evaluate classification accuracy on the FLORES+ *devtest* set. We show the language classification accuracy before and after LEACE transformation for the three models in Figure 8. After applying LEACE, classification accuracy for ID languages drops clearly, indicating successful erasure of linearly encoded language-specific information. Although the accuracy remains around 50–80%, this can be because the training and evaluation datasets are from

[3] https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct
[4] https://huggingface.co/google/gemma-3-4b-it
[5] https://huggingface.co/Qwen/Qwen3-4B
[6] https://huggingface.co/datasets/google/wmt24pp
[7] https://github.com/MicrosoftTranslator/NTREX/tree/468c6b69c7f6a75d31d4743d9daba2af566cc18d
[8] https://huggingface.co/datasets/openlanguagedata/flores_plus
[9] https://github.com/boschresearch/KLAR-CLC/tree/02f73c57ea17d4fd9f8b08fd55547058a601ba52
[10] https://huggingface.co/datasets/CohereLabs/Global-MMLU
[11] https://huggingface.co/datasets/facebook/belebele
[12] https://huggingface.co/datasets/google/xquad
[13] https://huggingface.co/datasets/CohereLabs/aya_dataset

Split / Benchmark	Languages
LEACE ID	English, French, Spanish, Mandarin Chinese, Russian, Japanese, Greek, Korean, Hebrew, Vietnamese, Bengali, Telugu
LEACE OOD	German, Turkish, Dutch, Ukrainian, Indonesian, Swahili
Evaluation: KLAR	(English), French, Spanish, Mandarin Chinese, Japanese, Korean, Hebrew, Greek, (Vietnamese only in §7)
Evaluation: GMMLU/BELEBELE	(English), French, Spanish, Mandarin Chinese, Japanese, Korean, Hebrew, Vietnamese, Bengali, Telugu
Evaluation: XQuAD	(English), Spanish, Mandarin Chinese, Vietnamese, Russian, Greek
Training: ID	English, French, Spanish, Mandarin Chinese, Japanese
Training: OOD	(KLAR) Korean, Hebrew, Greek, Vietnamese (GMMLU/BELEBELE) Korean, Hebrew, Vietnamese, Bengali, Telugu
Parameter Search: GMMLU *dev*	(English), French, Spanish, Mandarin Chinese, Japanese

Table 4: Languages used in each experimental split and evaluation benchmark.

the same domain (i.e., FLORES+) while LEACE is trained on a different domain (i.e., WMT24++ and NTREX). Also, as the number of samples for training the classifier is limited, the classifier may overfit. For OOD languages, the classifier retains almost 100% accuracy even after applying LEACE, indicating residual language-specific information.

D Optimal Transport Plans

Figure 9 shows examples of the transport plans calculated in CAROT for Llama3.1-8B. The transport plans are sparse, as a result of the perplexity-based filtering (Equation 6), which can increase robustness of alignment by suppressing noise. We can also see that semantically corresponding tokens across languages are aligned (e.g., "(Tur)quie" and "Turkey"), which supports the effectiveness of CAROT in aligning semantic representations.

E Evaluation Protocols

This section describes the evaluation configurations for the steering and training experiments in §6 and §7. We use four datasets for evaluation: KLAR Continent, GMMLU, BELEBELE, and XQuAD. We select these datasets because they provide parallel translations and are neutral in terms of language and culture, allowing fair evaluation of multilingual performance. For KLAR, five prompt templates in each language are provided in the official repository, and we use all of them for evaluation, resulting in 1,060 samples per language. For GMMLU, BELEBELE, and XQuAD, we prepare one prompt template in English and translate it into other languages with ChatGPT[14]. We randomly sample 30 instances for each GMMLU category and language to reduce the computational cost, resulting in 1,710 samples per language. BELEBELE has 900 samples per language, and XQuAD has 1,190 samples per language. For KLAR, GMMLU, and BELEBELE, we use *test* sets for evaluation, while for XQuAD, we use the *validation* set, since they only provide the *validation* set.

We let the model generate the answer with greedy decoding up to 50 tokens for KLAR and XQuAD, 500 tokens for GMMLU, and 100 tokens for BELEBELE, expecting that the reasoning tasks will require more tokens. We extract the final answer from the generated text in a rule-based manner based on the instructions in the prompt templates.

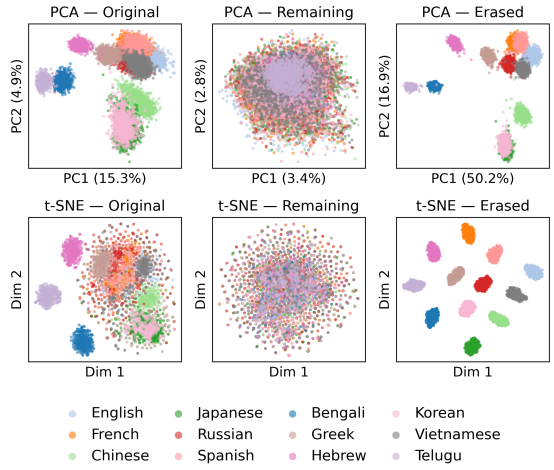
For hyperparameter tuning, we use the *dev* set of GMMLU (randomly sampled 50 instances per language) and perform a grid search. Regarding the parameters for Sinkhorn algorithm to solve optimal transport in CAROT (Equation 5), we set the regularization parameter ϵ_{OT} to 0.05 and the unbalanced mass penalty ρ to 0.5.

F Baseline Steering Methods

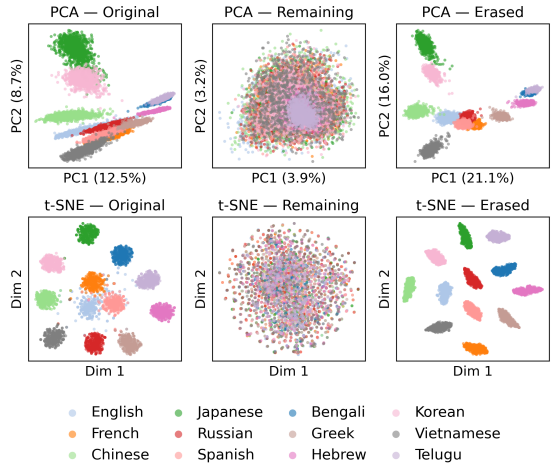
In the steering experiments (§6), we compare CAROT with two baseline methods: *Lang Vec* and *MLRS*.

Lang Vec adds the difference between the mean hidden vectors of the source and target languages to the hidden states at a specific layer to bring the representation of the target language closer to that

[14] https://chatgpt.com/overview/

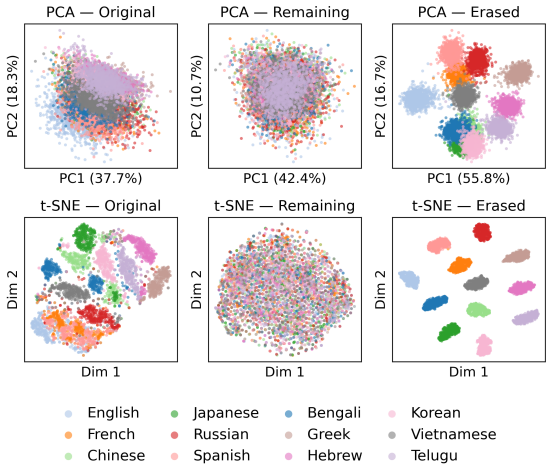


(a) Layer 6

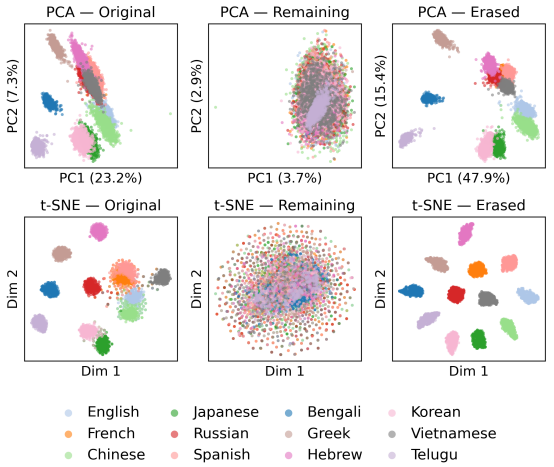


(b) Layer 25

Figure 6: Scatter plot of original (left), remaining (center), and LEACE-erased (right) representations of Llama3.1-8B at layer 6 and 25. Each point represents the representation of a sentence, colored by language.



(a) gemma3-4b layer 6



(b) Qwen3-4B layer 7

Figure 7: Scatter plot of original (left), remaining (center), and LEACE-erased (right) representations of gemma3-4b and Qwen3-4B at layers 6 and 7. Each point represents the representation of a sentence, colored by language.

of the source language:

$$\mathbf{h}_{\text{tgt}}^{\text{new}} = \mathbf{h}_{\text{tgt}} + \alpha(\bar{\mathbf{h}}_{\text{src}} - \bar{\mathbf{h}}_{\text{tgt}}). \quad (11)$$

We calculate the mean vectors on the sentence representations (Equation 1) of the same parallel datasets as used for training LEACE (§4). α is a hyperparameter that controls the strength of the steering. We search α from {0.5, 1.0, 1.5, 2.0} and the layer from every three layers for each model.

For *MLRS* (Zhao et al., 2025a), it first calculates the mean hidden vectors of the last token activations of each language in the training data and concatenates them to form $\mathbf{M} \in \mathbb{R}^{d \times L}$, where L is the number of languages. Then it minimizes the following objective:

$$\min_{\mathbf{M}_a, \mathbf{M}_s, \Gamma} \|\mathbf{M} - \mathbf{M}_a \mathbf{1}^\top - \mathbf{M}_s \Gamma\|_F^2 \quad (12)$$

$$\text{s.t. } \text{Span}(\mathbf{M}_a) \perp \text{Span}(\mathbf{M}_s), \quad (13)$$

where $\mathbf{M}_a \in \mathbb{R}^{d \times 1}$ is the language-agnostic component, $\mathbf{M}_s \in \mathbb{R}^{d \times r}$ is the language-specific component, and $\Gamma \in \mathbb{R}^{r \times L}$ is the coefficient matrix. Following the paper that MLRS is based on (Xie et al., 2022), we set $r = L - 1$. The optimal solution of this objective can be obtained using singular value decomposition. At inference time, MLRS intervenes in the language-specific component of the hidden states as follows:

$$\mathbf{h}_{\text{tgt}}^{\text{new}} = \mathbf{h}_{\text{tgt}} - \lambda \mathbf{M}_s \mathbf{M}_s^\top \mathbf{h}_{\text{tgt}}, \quad (14)$$

where λ is a hyperparameter that controls the direction and strength of the intervention. We follow the original paper and set $\lambda > 0$ (suppressing language-specific information) from the one-third to two-thirds layers and $\lambda < 0$ (adding back language-specific information) for the last one-third layers. We search λ_{sup} from {0.1, 0.2, 0.3, 0.4} for suppressing and λ_{add} from {-0.1, -0.2, -0.3, -0.4} for adding back.

G Selected Hyperparameters

In the steering experiments (§6) and training experiments (§7), we select the best hyperparameters for each method based on the evaluation accuracy on the GMMLU *dev* set. The selected hyperparameters for steering methods are shown in Table 5, and those for training methods are shown in Table 6. Note that for the layer and θ of CAROT in the training experiments, we use the best values selected in the steering experiments.

H Additional Steering Results

Agreement Ratio of Steering Results. We show the agreement ratio (i.e., the percentage of instances for which the model's answer in the target language matches that in the source language) of each steering method in Table 7, along with the accuracy reported in Table 1. The agreement ratio directly measures CLA, and a higher agreement ratio indicates that the model is more consistent in its answers across languages. Overall, the agreement ratio correlates with the accuracy, and CAROT achieves the highest agreement ratio in most cases. Moreover, CAROT tends to increase the agreement ratio more than the accuracy (e.g., for Llama3.1-8B on GMMLU, the accuracy increases by 1.4 points while the agreement ratio increases by 2.4 points). This indicates that the increase in the accuracy of CAROT is due to the improvement in CLA.

Detailed Layer-wise and Language-wise Steering Results. We present the layer-wise per-language accuracy and fidelity when applying steering with CAROT, calculated during the hyperparameter search on GMMLU dev. The results for Llama3.1-8B are shown in Figure 4, and the results for gemma3-4b and Qwen3-4B are shown in Figure 10. CAROT improves accuracy across languages, especially in the early layers, and maintains high fidelity in most cases. However, for gemma3-4b, the fidelity is low in some languages (e.g., Chinese and Japanese). This can be because gemma3-4b originally (without steering) tends to generate English reasoning steps even when prompted in other languages.

Furthermore, we show the layer-wise and per-language results of steering with Lang Vec on GMMLU dev in Figure 11. The accuracy is less improved relative to CAROT, and the fidelity often becomes low in early-to-middle layers. Since Lang Vec adds the diff-in-means language vector to the hidden states of the target language, it can remove language information and thus change the output language, which can cause low fidelity. On the other hand, CAROT explicitly preserves the language representations, which can contribute to high fidelity.

Ablation of Each Component of CAROT. We conduct an ablation study to analyze the contribution of each component of CAROT, including LEACE-based decomposition, perplexity-based filtering of the OT plan, and the token-level OT alignment. Table 8 compares the evaluation accuracy and the fidelity of CAROT, CAROT without LEACE-based decomposition (*No LEACE*), CAROT without filtering the OT plan (*No Filter*), and swapping only the language-agnostic representations of the last token between the inputs in source and target languages (*Last Token*). Last Token corresponds to CAROT without token-level OT alignment. Note that No Filter is included in the hyperparameter search for CAROT, which corresponds to $\theta = 1.0$.

The results show that CAROT is most stable across models and benchmarks. CAROT outperforms No LEACE and Last Token in 9 and 10 out of 12 cases, respectively, in terms of accuracy. Although the accuracies of No Filter are comparable to those of CAROT, No Filter is sometimes unstable (e.g., for Qwen3-4B on KLAR, the accuracy is 14.0 points lower than CAROT), and the fidelity is often lower than CAROT (9 out of 12 cases). These results suggest that LEACE-based decomposition and filtering of the OT plan make CAROT more robust, and that token-level OT alignment is effective in improving accuracy.

I Training Details

MidAlign. Here, we provide the detailed specification of the training algorithms for the baseline method, MidAlign. MidAlign fine-tunes the model by alternating between a contrastive step and a language-modeling (LM) step. In the contrastive

Method	Llama3.1-8B	gemma3-4b	Qwen3-4B
CAROT	{layer: 12, θ: 0.25}	{layer: 6, θ: 0.5}	{layer: 9, θ: 0.75}
Lang Vec	{layer: 21, α: 0.5}	{layer: 33, α: 0.5}	{layer: 27, α: 2.0}
MLRS	{λ_{sup}: 0.1, λ_{add}: -0.4}	{λ_{sup}: 0.1, λ_{add}: -0.4}	{λ_{sup}: 0.1, λ_{add}: -0.4}

Table 5: Selected hyperparameters for each steering method and model.

Method	Llama3.1-8B	gemma3-4b	Qwen3-4B
CAROT	{lr: 5e-5}	{lr: 1e-4}	{lr: 1e-4}
LM	{lr: 1e-5}	{lr: 1e-5}	{lr: 5e-5}
MidAlign	{lr: 5e-5, τ: 0.1}	{lr: 1e-4, τ: 0.1}	{lr: 1e-4, τ: 0.1}

Table 6: Selected hyperparameters for each training method and model.

		GMMLU		BELEBELE	
Model	Method	Acc	Agr	Acc	Agr
Llama3.1-8B	Unsteered	48.4	51.5	74.7	75.3
	Lang Vec	43.8	47.1	74.5	75.2
	MLRS	43.7	47.0	77.5	78.2
	CAROT	**49.8**	**53.9**	**81.9**	**83.9**
gemma3-4b	Unsteered	49.4	57.0	64.2	64.1
	Lang Vec	49.0	56.9	63.8	63.3
	MLRS	48.2	56.2	64.8	63.8
	CAROT	**51.8**	**61.4**	**66.0**	**65.6**
Qwen3-4B	Unsteered	56.8	61.2	62.3	62.7
	Lang Vec	54.9	59.0	63.3	63.7
	MLRS	55.9	60.1	**64.9**	**65.2**
	CAROT	**59.6**	**64.2**	62.5	63.1

Table 7: Agreement ratio (Agr, %) alongside accuracy (Acc, %) on the two multiple-choice datasets (GMMLU and BELEBELE). The two metrics induce the same ranking over methods in five of six model–dataset settings.

step, at the specified layer l, the source and target hidden states are calculated by mean-pooling hidden vectors over non-padding tokens, giving $\mathbf{h}_s^{(l)}, \mathbf{h}_t^{(l)} \in \mathbb{R}^{B\times d}$, where B is the batch size and d is the hidden dimension. The symmetric InfoNCE loss is then computed as:

$$\mathcal{L}^{(l)}_{\text{NCE, col}} = -\log \frac{\exp(\text{sim}(\mathbf{h}_s^{(l)}, \mathbf{h}_t^{(l)})/\tau)}{\sum_{i=1}^{B} \exp(\text{sim}(\mathbf{h}_s^{(l)}, \mathbf{h}_{t,i}^{(l)})/\tau)}, \tag{15}$$

$$\mathcal{L}^{(l)}_{\text{NCE, row}} = -\log \frac{\exp(\text{sim}(\mathbf{h}_t^{(l)}, \mathbf{h}_s^{(l)})/\tau)}{\sum_{i=1}^{B} \exp(\text{sim}(\mathbf{h}_t^{(l)}, \mathbf{h}_{s,i}^{(l)})/\tau)}, \tag{16}$$

$$\mathcal{L}^{(l)}_{\text{contrastive}} = \frac{1}{2}(\mathcal{L}^{(l)}_{\text{NCE, col}} + \mathcal{L}^{(l)}_{\text{NCE, row}}) \tag{17}$$

where τ is the temperature and $\text{sim}(\cdot,\cdot)$ is the cosine similarity function. As in the original paper, we use the alignment datasets for the contrastive step (parallel dataset) and the task-specific dataset for the LM step.

Training Parameters. We train all models for one epoch with a batch size of 16 using the AdamW optimizer. The learning rate is searched from {5e-4, 1e-4, 5e-5, 1e-5} for each model and method. For MidAlign, we search the temperature τ from {0.1, 0.5, 1.0}. The selected hyperparameters are shown in Table 6. For other parameters, we set weight decay to 0.01 and adopt a linear learning rate scheduler with 300 warmup steps. We apply LoRA to all linear modules with rank $r = 64$, scaling factor $\alpha = 64$, and a dropout rate of 0.05.

We limit the maximum number of instruction-tuning training samples to 2,000 per language to balance the number of training data across languages, and we sample the same number of parallel sentences for the alignment step. We set the maximum input length to 500 tokens for task-specific data and 64 tokens for parallel sentences. As for the training layers for MidAlign, we select layers 16, 17, and 18 for Llama3.1-8B, gemma3-4b, and Qwen3-4B, respectively, following the analysis in the original paper, which shows that the middle layers are most effective for MidAlign.

Analysis of Training. To further analyze the training results, we calculate the cosine similarity between the source and target language-agnostic representations at each layer before and after training. The results for Llama3.1-8B are shown in Figure 12. We observe that the similarity increases most significantly around the training layer (layer 12 for Llama3.1-8B) after training with CAROT, which indicates that the training successfully encourages the model to produce language-agnostic representations that are close to the source representations. The increase in similarity lasts across

Model	Method	GMMLU (9)		KLAR (7)		BELEBELE (9)		XQuAD (5)	
		Acc	Fid	Acc	Fid	Acc	Fid	Acc	Fid
Llama3.1-8B	CAROT	49.8	93.1	73.3	93.8	81.9	85.3	**41.2**	72.7
	No LEACE	**55.7**	91.3	**73.6**	92.5	81.4	82.7	39.6	69.9
	No Filter	55.5	91.7	71.9	91.6	**82.1**	85.5	34.7	71.3
	Last Token	49.5	93.5	67.8	93.6	76.5	88.9	36.2	78.0
gemma3-4b	CAROT	51.8	62.6	75.8	96.4	**66.0**	92.9	26.0	82.5
	No LEACE	49.4	63.2	67.9	92.5	64.1	94.1	21.2	86.2
	No Filter	**53.2**	59.0	**77.2**	93.3	65.1	87.9	**31.9**	75.8
	Last Token	49.4	63.6	67.7	94.5	64.7	93.8	22.0	86.2
Qwen3-4B	CAROT	59.6	98.1	**68.0**	90.4	62.5	89.5	45.6	69.9
	No LEACE	57.2	98.3	60.4	92.2	63.6	89.7	44.7	71.7
	No Filter	**59.8**	97.3	54.0	87.2	57.8	91.5	36.8	71.2
	Last Token	57.0	98.4	56.8	93.9	**64.7**	89.5	**46.1**	71.5

Table 8: Ablation results of steering with CAROT, CAROT without LEACE-based decomposition (No LEACE), CAROT without perplexity-based filtering of the OT plan (No Filter), and swapping the language-agnostic representations of only the last token between the source and target languages (Last Token). Bold marks the highest value per metric group.

layers, which can contribute to the improved performance in the evaluation benchmarks.

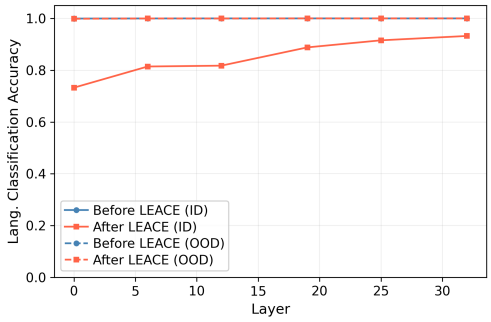
J Computational Resources

All the experiments are conducted on a single NVIDIA H100 GPU. Hyperparameter search of the steering experiments in §6 took around 2 hours for each model and method. Hyperparameter search of the training experiments in §7 took around 1.5 hours for each model for LM and CAROT, and around 4 hours for MidAlign, since we search the temperature τ in addition to the learning rate for MidAlign. For the evaluation of each model and method, it took 2.5 hours for GMMLU, 20 minutes for KLAR, BELEBELE, and XQuAD, respectively.

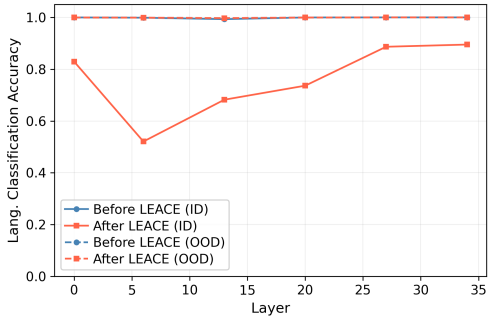
K LLM Usage

We used Claude Code[15] as a coding assistant for implementing the scripts for the experiments and visualizations. We also used ChatGPT to proofread the paper and to translate the prompt template used in evaluation into different languages.

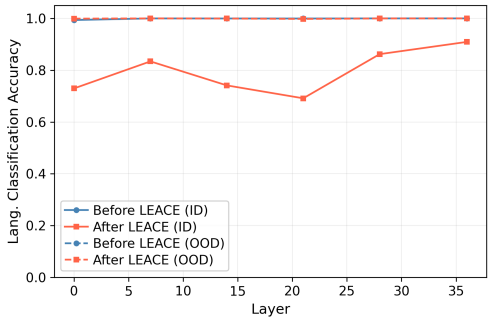
[15] https://code.claude.com/docs/en/overview



(a) Llama3.1-8B

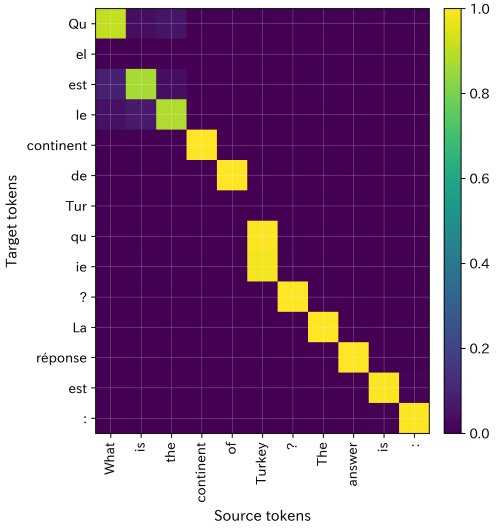


(b) gemma3-4b

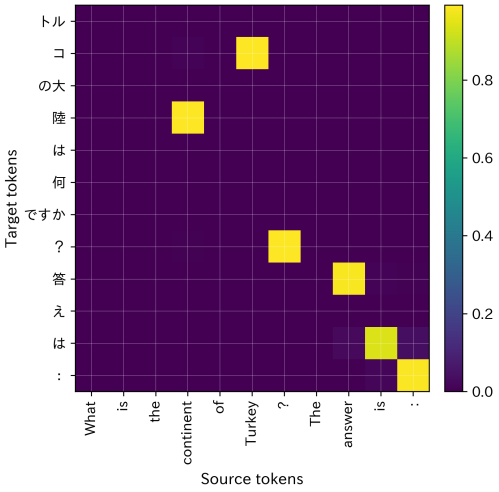


(c) Qwen3-4B

Figure 8: Language classification accuracy before (blue) and after (red) LEACE transformation at every six layers of Llama3.1-8B, gemma3-4b and Qwen3-4B. We report the accuracies of in-distribution (solid) and out-of-distribution (dashed) languages separately. The accuracies for ID languages before LEACE and for OOD languages before and after LEACE are almost 100%.

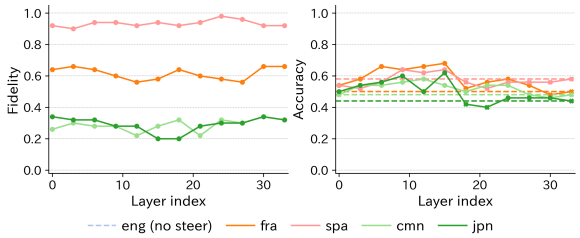


(a) French at layer 12

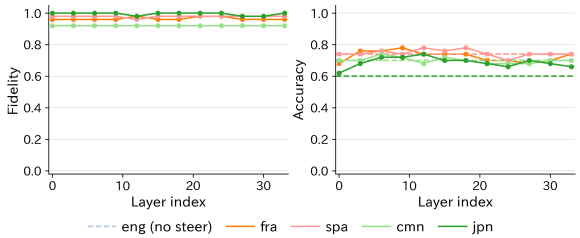


(b) Japanese at layer 12

Figure 9: Visualization of the calculated OT transport plan $\mathbf{T}$ on Llama3.1-8B for a sample sentence pair in French-English (a) and Japanese-English (b). The brightness of each cell corresponds to the value of T_{ij}, indicating how much the j-th token in the source language (English) is transported to the i-th token in the target language (French or Japanese).

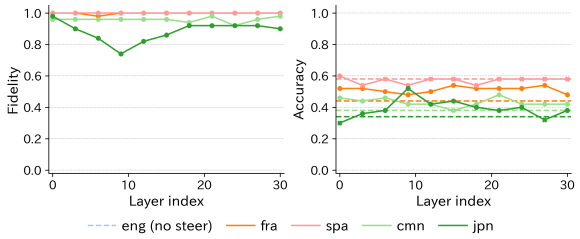


(a) gemma3-4b

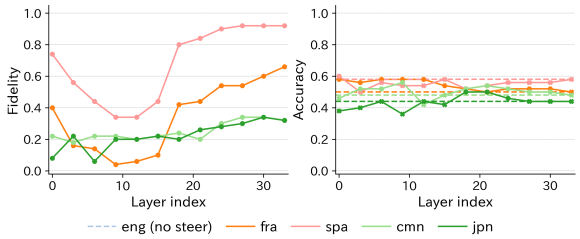


(b) Qwen3-4B

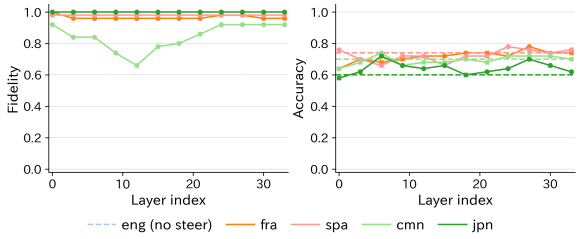
Figure 10: Fidelity (left) and accuracy (right) of gemma3-4b and Qwen3-4B on GMMLU dev when applying steering with CAROT. The results of Llama3.1-8B are shown in Figure 4.



(a) Llama3.1-8B



(b) gemma3-4b



(c) Qwen3-4B

Figure 11: Fidelity (left) and accuracy (right) of Llama3.1, gemma3-4b and Qwen3-4B on GMMLU dev when applying steering with **Lang Vec**.

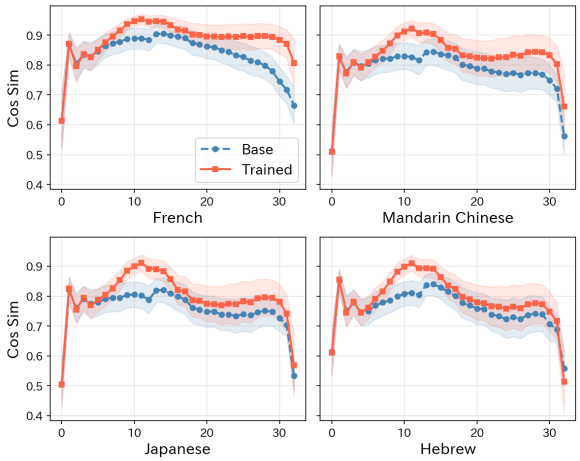


Figure 12: Cosine similarity between the source and target language-agnostic representations of Llama3.1-8B across layers before and after training with CAROT, calculated on FLORES+ *devtest* set.